COMPASS Synoptic: Porewater SO4/Cl

July 2026 Samples

2026-11-02

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1 Run Information

Run date: 2026-07-30
Run by: Zoe Read
Run notes: Run notes

SO4 file used: CO2xN_202605_SO4.txt

Cl file used: CO2xN_202605_Cl.txt

Changed sample names:

Samples removed:

```
## [1] "5" "6"
```

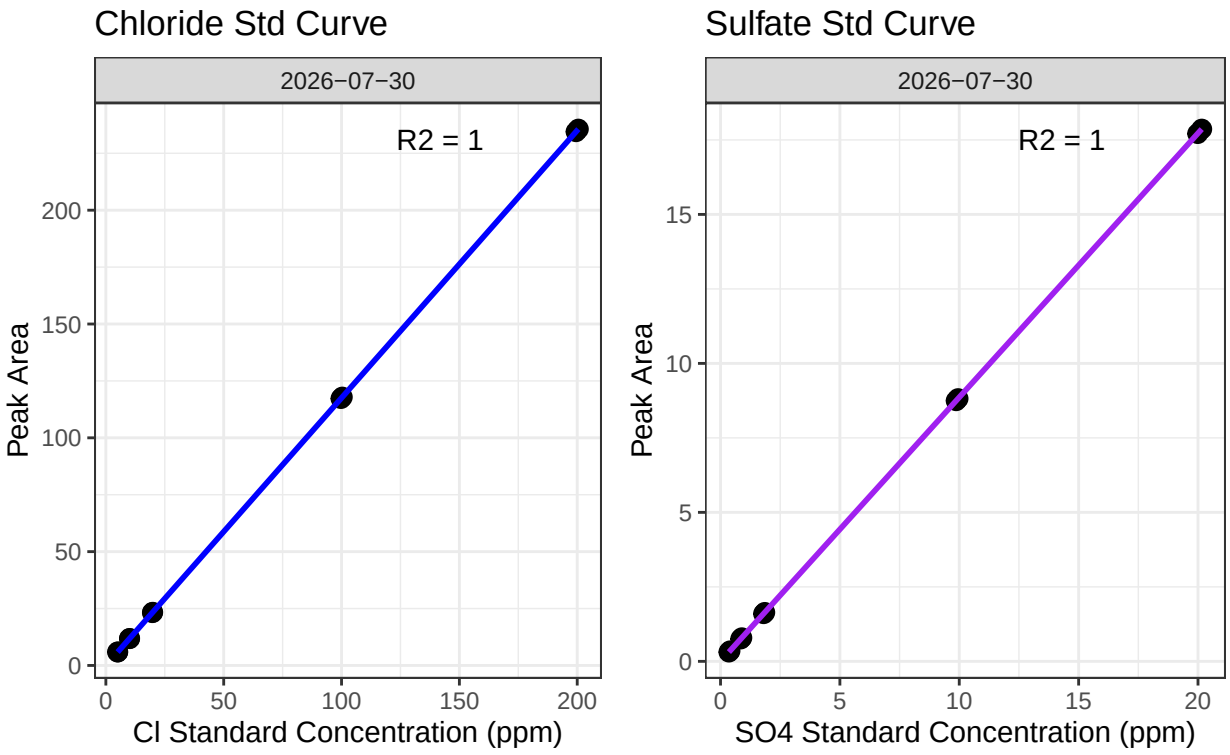
Links:

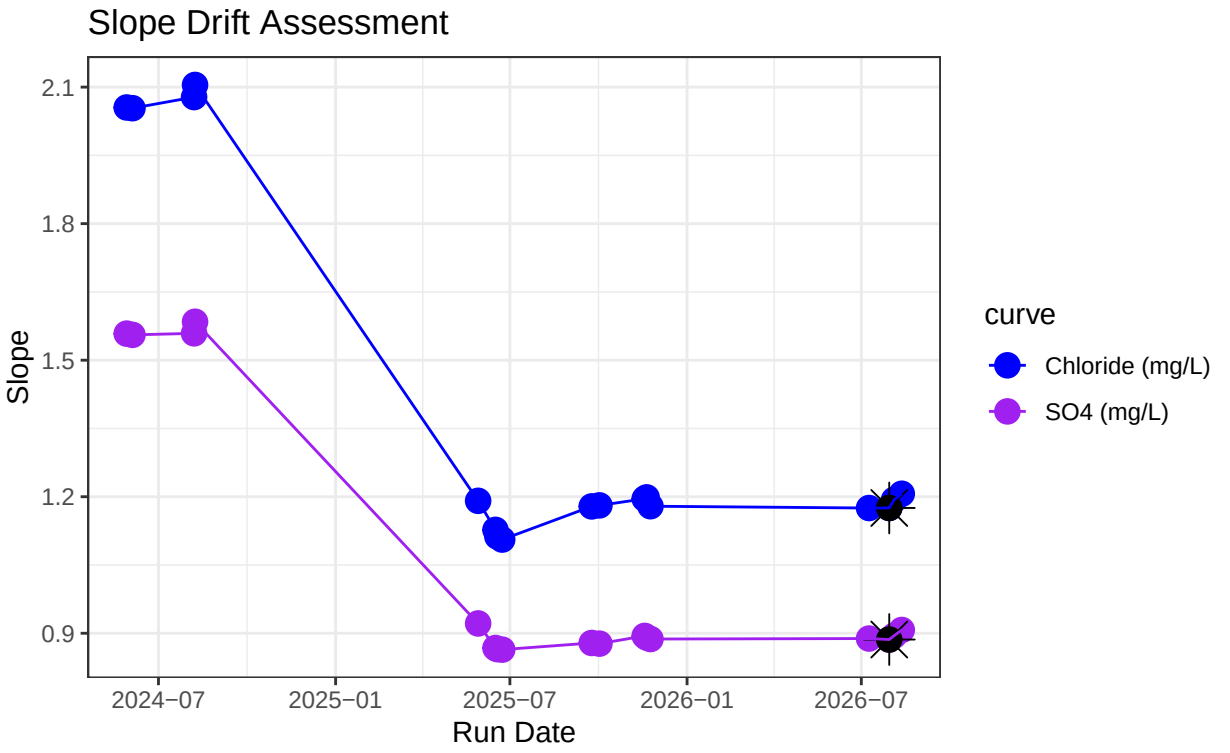
Base QAQC Code Template Used

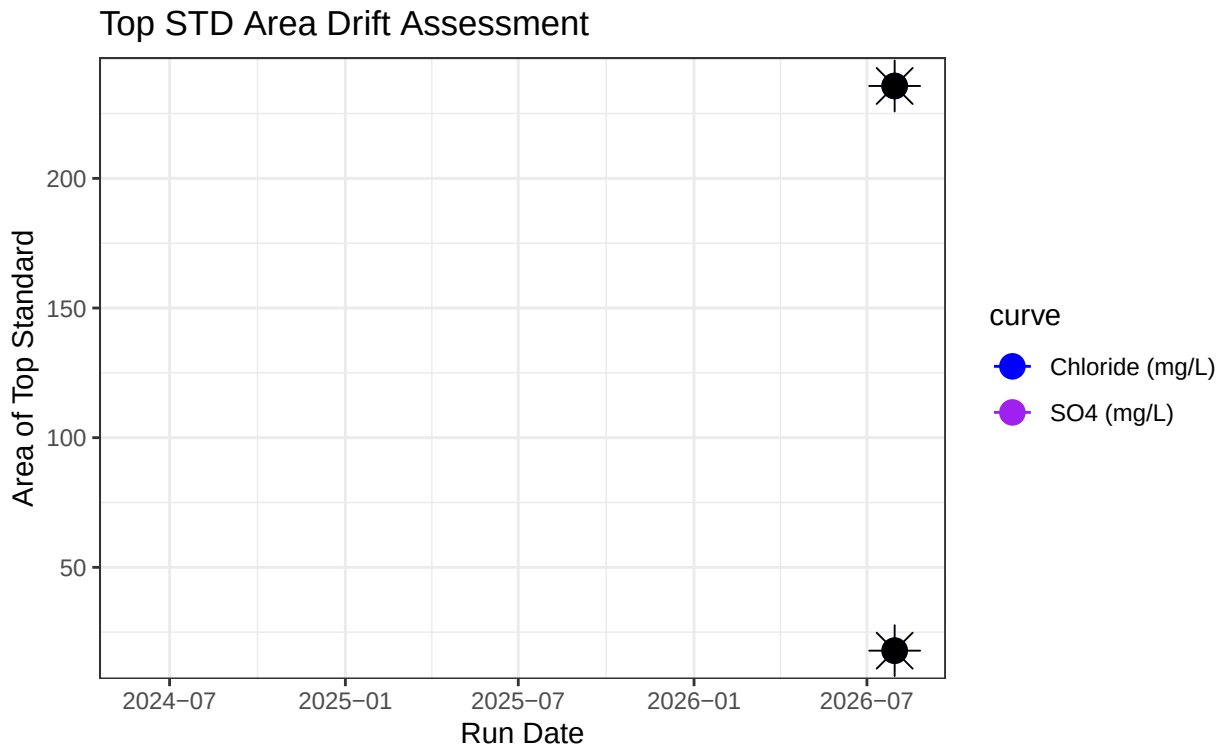
Project-Specific QAQC Code Template Used

Protocol

1.1 Assess Standard Curves








```
## [1] "PROCEED - Cl Curve r2 GOOD"
```

```
## [1] "PROCEED - SO4 Curve r2 GOOD"
```

run_date	analyte	slope_flag	intercept_flag
2026-07-30	Cl	PROCEED - Slope is within 2 sd of previous slopes	PROCEED - Intercept is within 2 sd of previous intercepts
2026-07-30	SO4	PROCEED - Slope is within 2 sd of previous slopes	PROCEED - Intercept is within 2 sd of previous intercepts

1.2 Assess Check Standards

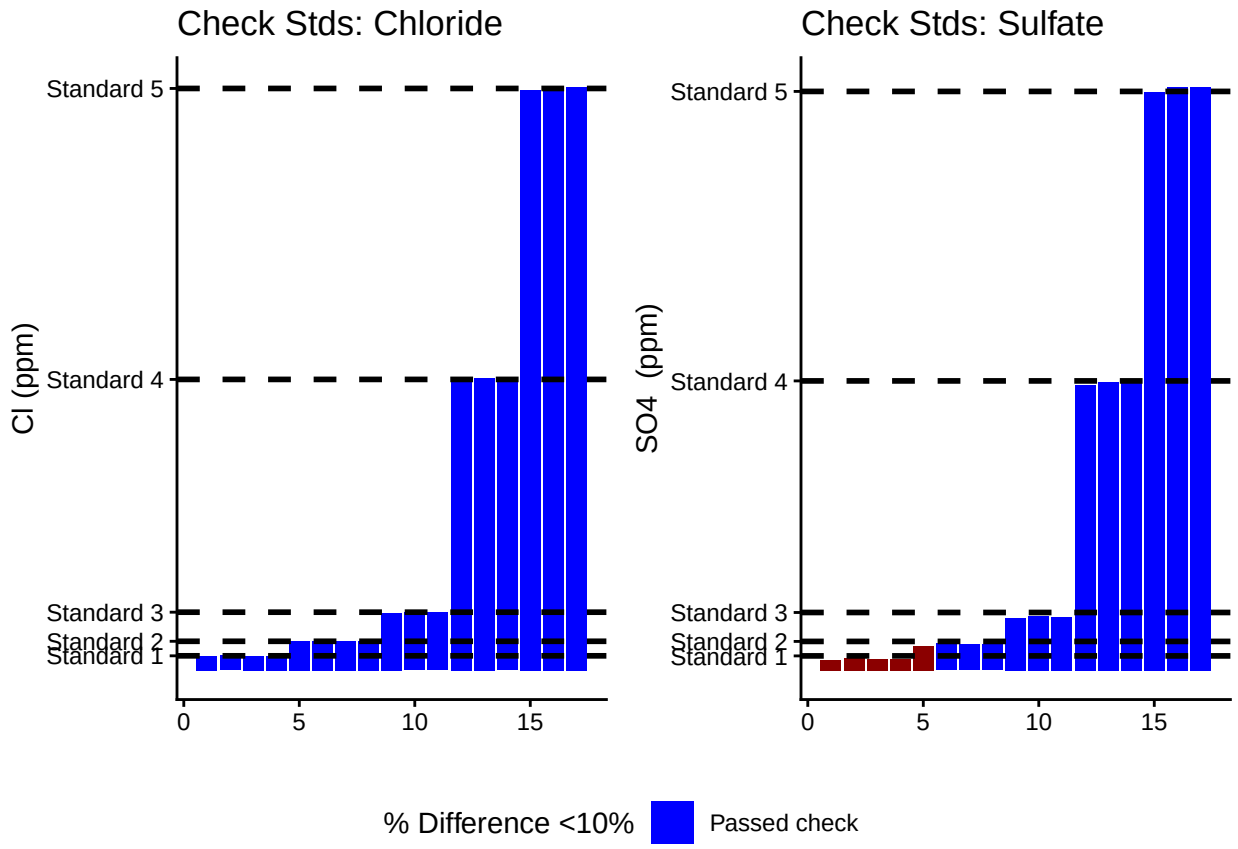
sample_ID	mean_Cl	cv_Cl	flag_Cl
Standard 1	5.056	0.007	PROCEED - Chloride Check Standard RSD within Range
Standard 2	10.080	0.005	PROCEED - Chloride Check Standard RSD within Range
Standard 3	19.837	0.004	PROCEED - Chloride Check Standard RSD within Range
Standard 4	100.187	0.004	PROCEED - Chloride Check Standard RSD within Range
Standard 5	200.178	0.003	PROCEED - Chloride Check Standard RSD within Range

sample_ID	mean_SO4	cv_SO4	flag_SO4
Standard 1	0.392	0.078	PROCEED - Sulfate Check Standard RSD within Range
Standard 2	0.891	0.038	PROCEED - Sulfate Check Standard RSD within Range

sample_ID	mean_SO4	cv_SO4	flag_SO4
Standard 3	1.844	0.020	PROCEED - Sulfate Check Standard RSD within Range
Standard 4	9.931	0.005	PROCEED - Sulfate Check Standard RSD within Range
Standard 5	20.096	0.005	PROCEED - Sulfate Check Standard RSD within Range

```
## [1] "PROCEED - >80% of Chloride Check Standards have RSD within range"
```

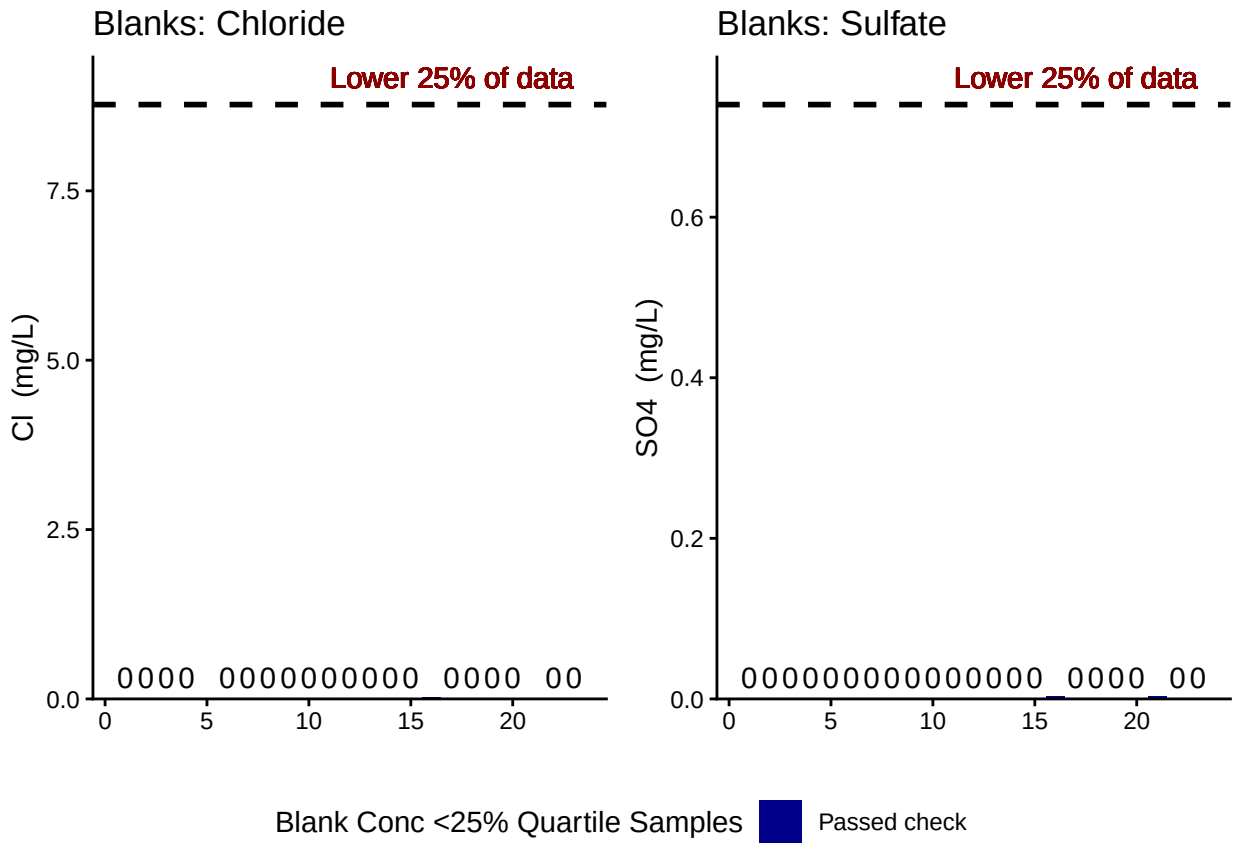
```
## [1] "PROCEED - >80% of Sulfate Check Standards have RSD within range"
```




```
## [1] "PROCEED - >80% of Chloride Check Standards are within range of expected concentration"
```

```
## [1] "REASSESS - <80% of Sulfate Check Standards are within range of expected concentration"
```

1.3 Assess Blanks

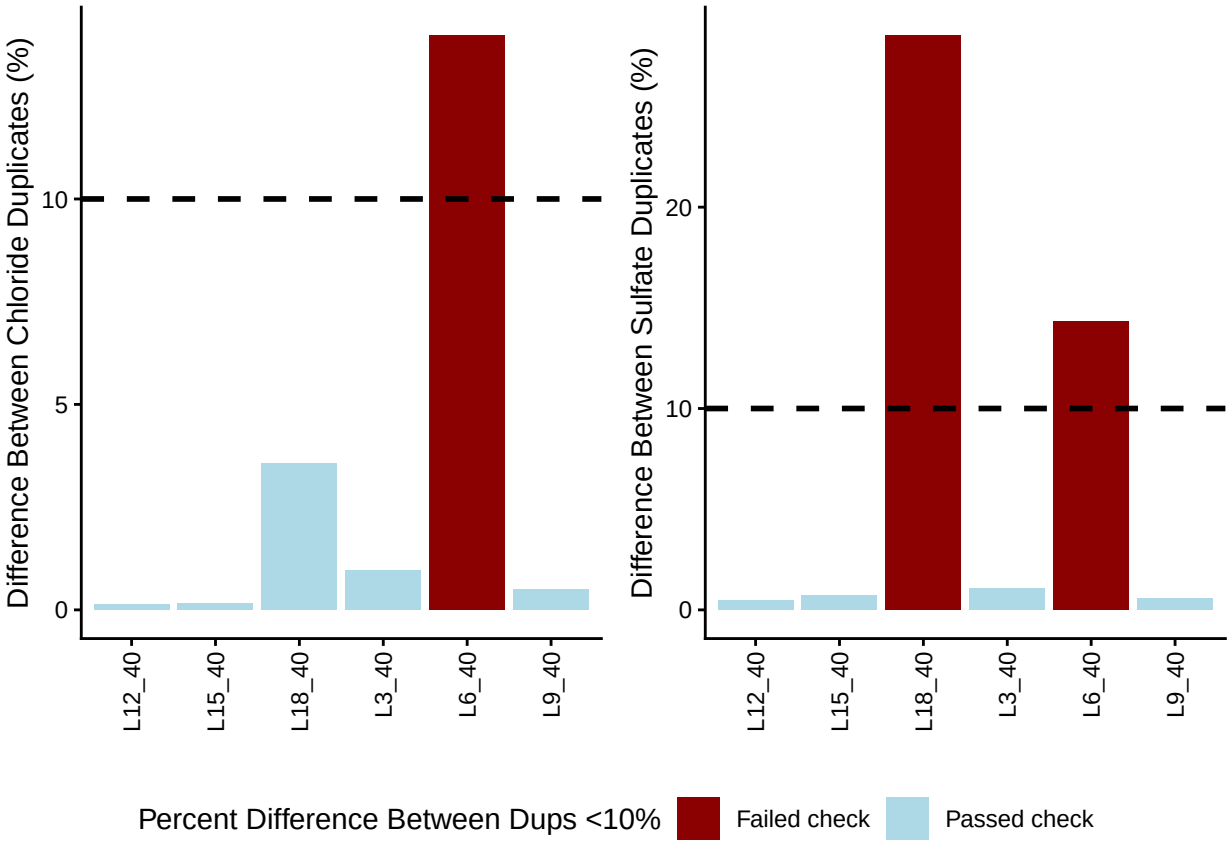


analyte	run_date	mean_blanks
Cl	2026-07-30	0.002
SO4	2026-07-30	0.000

```
## [1] "PROCEED - >80% of Chloride Blank concentrations are in the lower 25% quartile of samples"
```

```
## [1] "PROCEED - >80% of Sulfate Blank concentrations are in the lower 25% quartile of samples"
```

1.4 Assess Duplicates

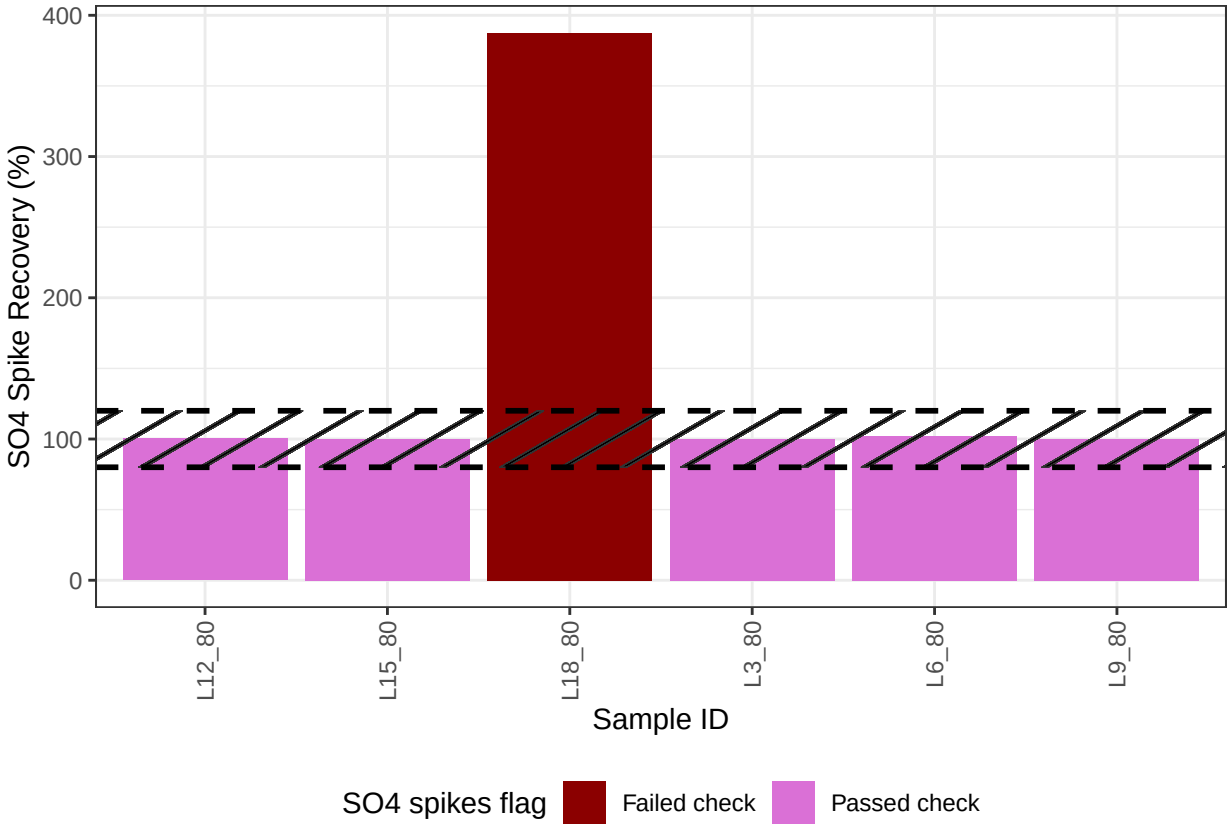



```
## [1] "PROCEED - >80% of Chloride Duplicates have a percent difference <10%"
```

```
## [1] "REASSESS - <80% of Sulfate Duplicates have a CV <10%"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

1.6 Assess SO4 Spikes




```
## [1] "PROCEED - >80% of SO4 spikes have a recovery between the high and low cutoff"
```

1.7 Check if samples are within the range of the standard curve

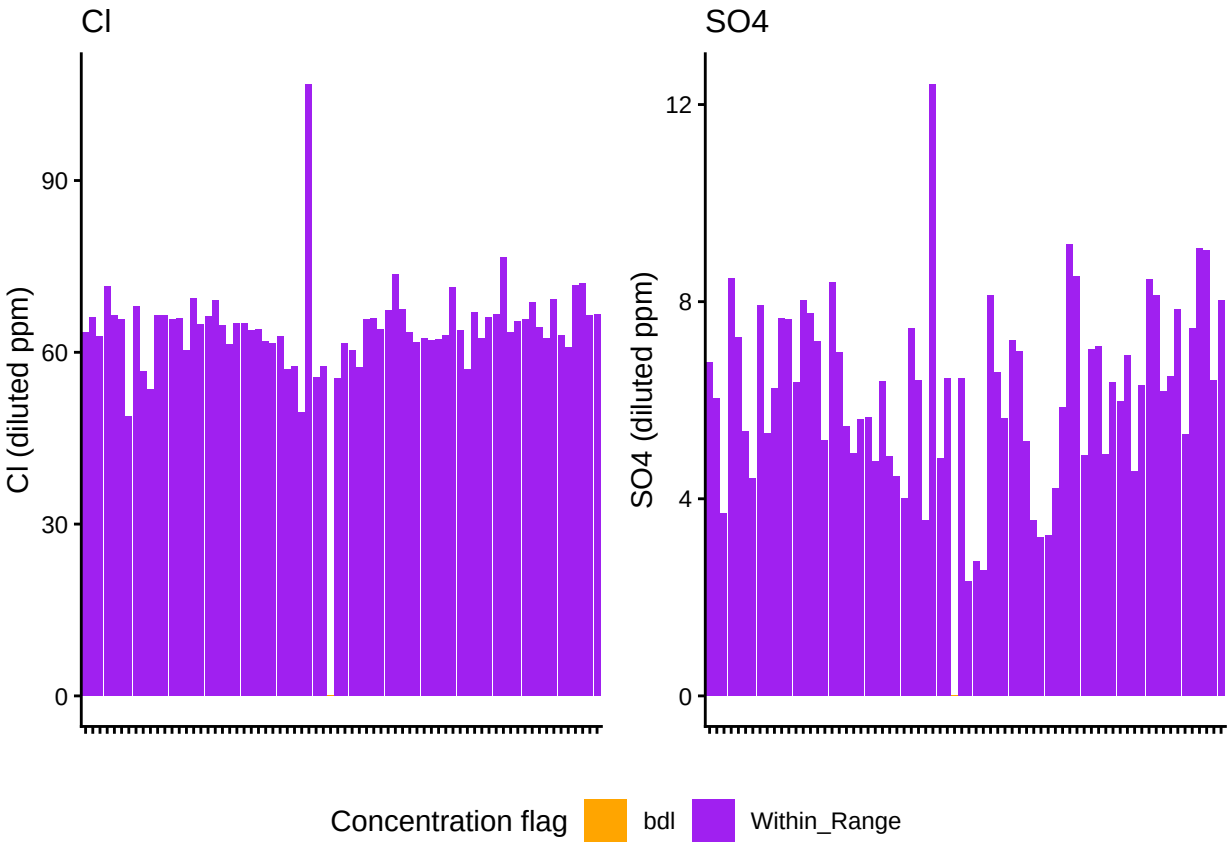
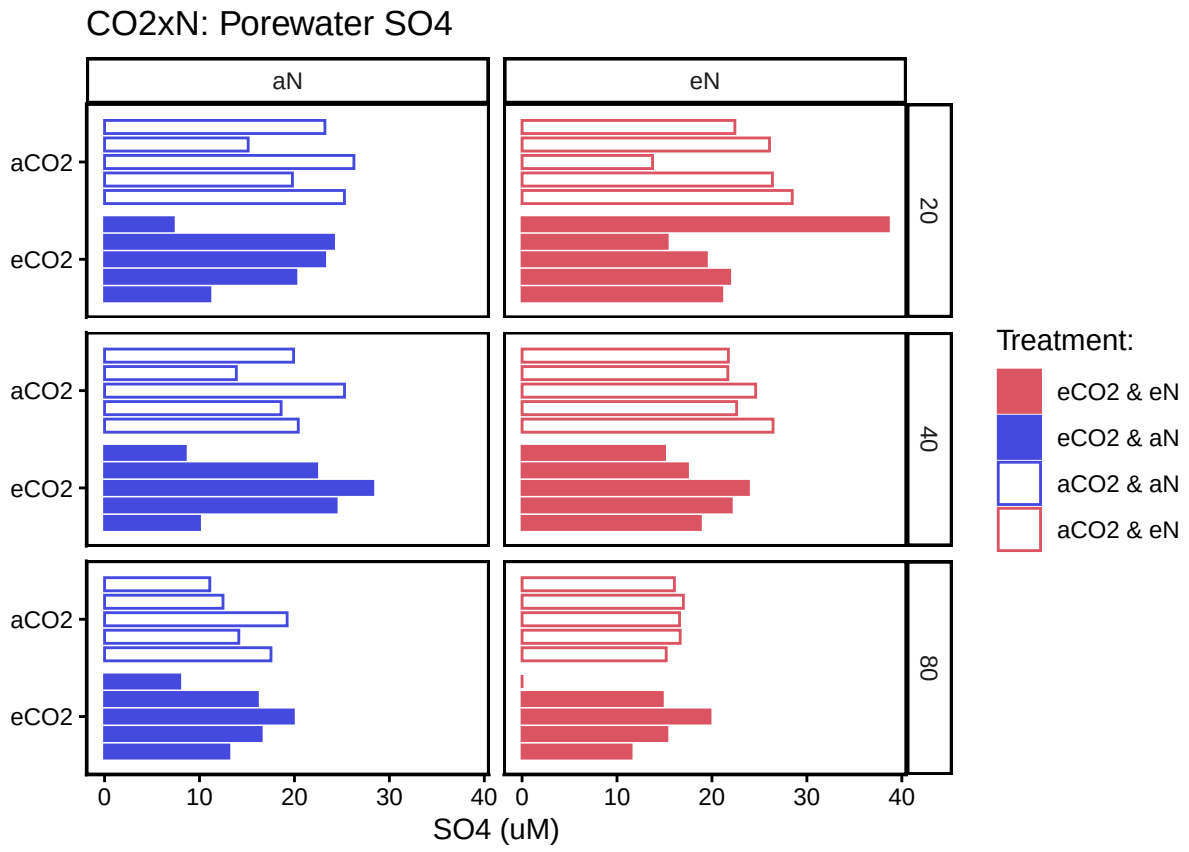


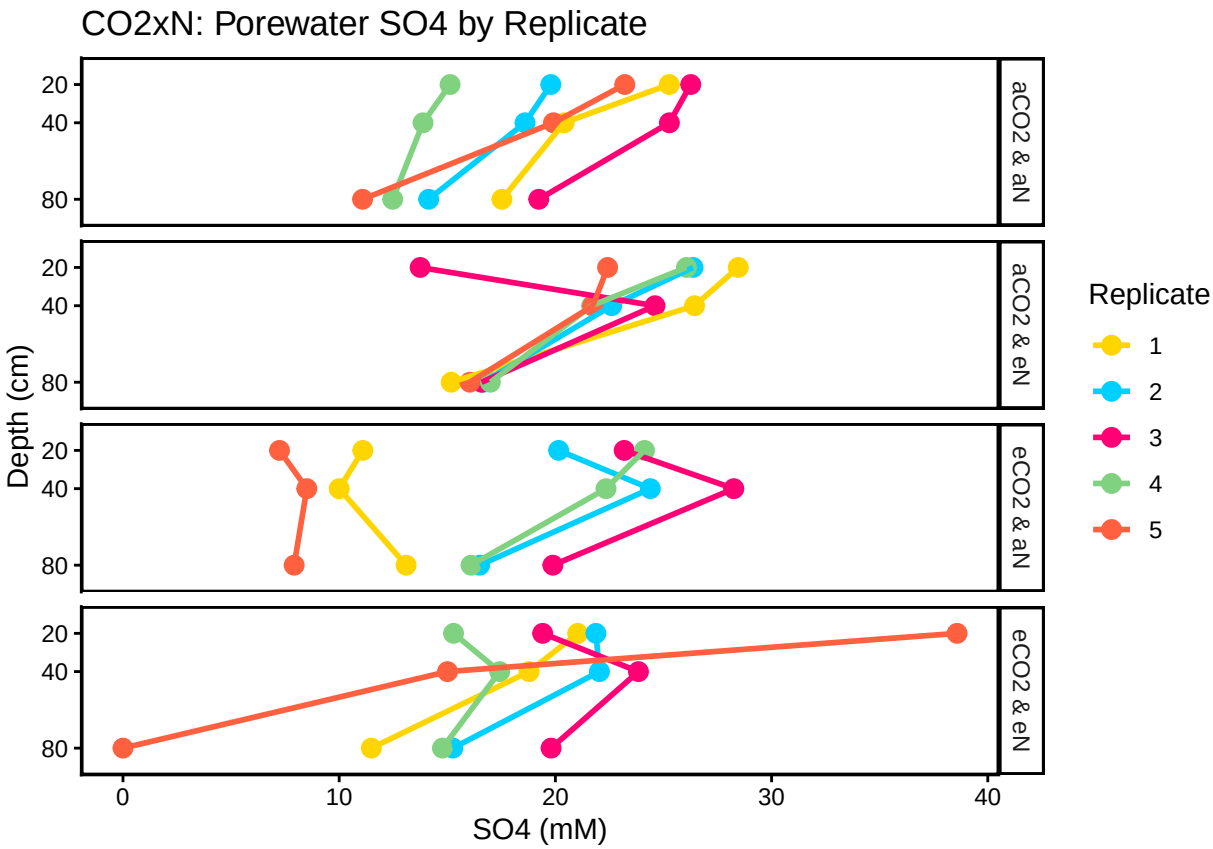
Table 5: Sample Concentration Flags

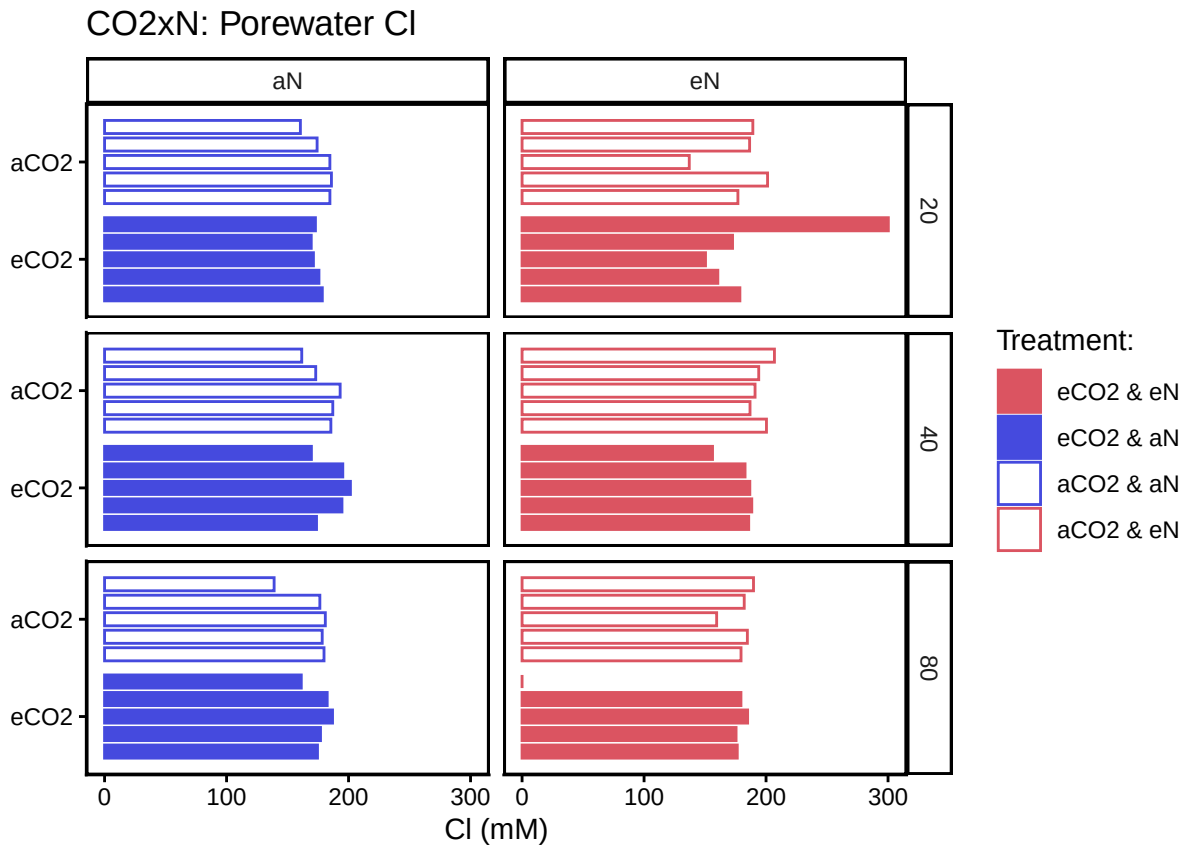
Conc_flag	Percent_samples_SO4	Percent_samples_Cl
bdl	1.388889	1.388889
Within_Range	98.611111	98.611111

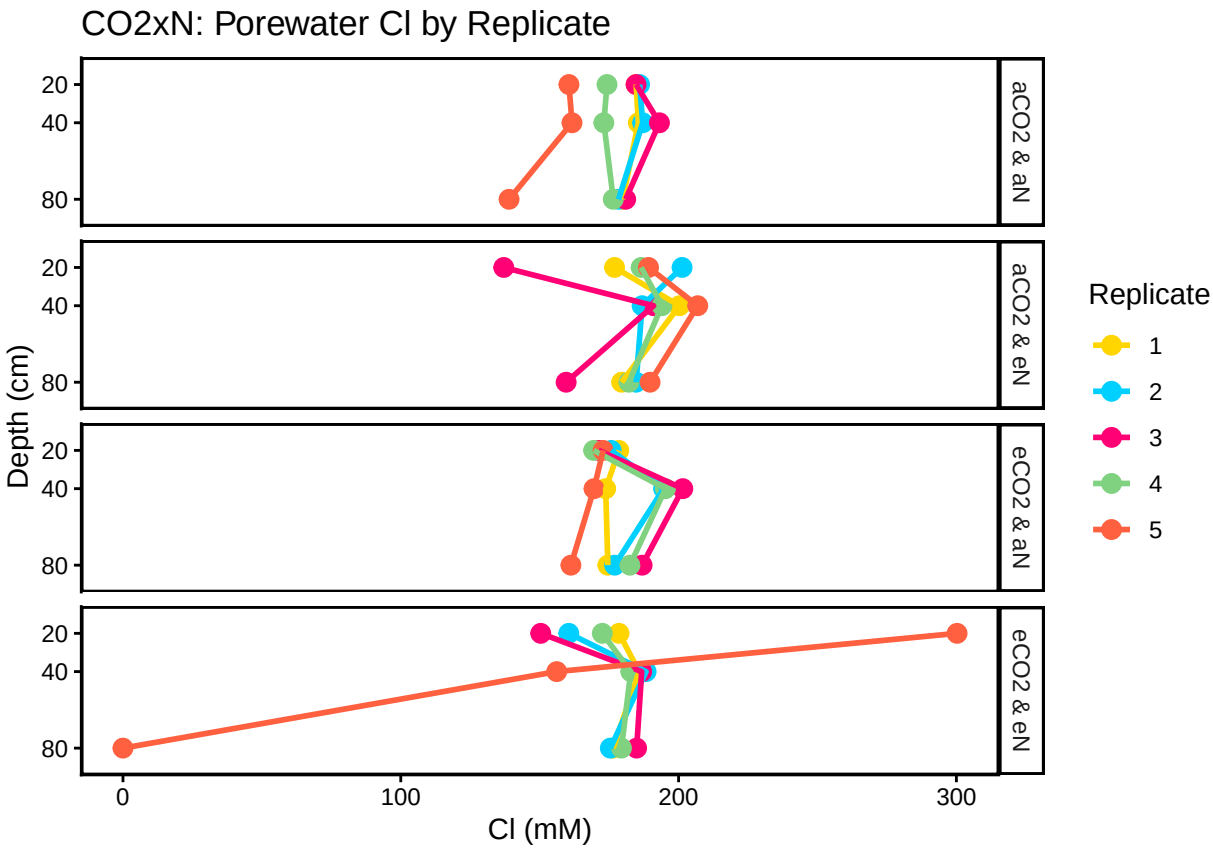
1.8 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```


CO2xN: Porewater SO4
aN
eN
aCO2
eCO2
20
40
80
0
10
20
30
40
SO4 (uM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN


CO2xN: Porewater SO4 by Replicate
Depth (cm)
20
40
80
aCO2 & aN
aCO2 & eN
eCO2 & aN
eCO2 & eN
0
10
20
30
40
SO4 (mM)
Replicate
1
2
3
4
5


CO2xN: Porewater Cl
aN
eN
aCO2
eCO2
20
40
80
0
100
200
300
Cl (mM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN


CO2xN: Porewater Cl by Replicate
Depth (cm)
20
40
80
aCO2 & aN
aCO2 & eN
eCO2 & aN
eCO2 & eN
0
100
200
300
Cl (mM)
Replicate
1
2
3
4
5

#end